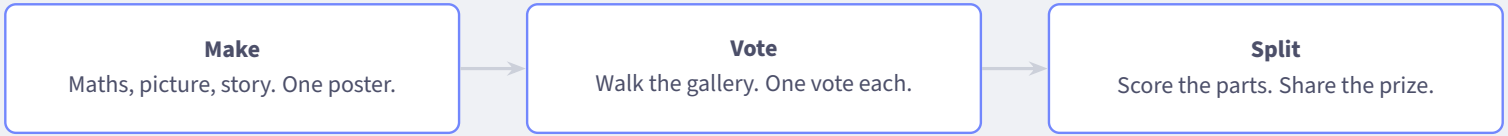
# Cooperative games

Three people made one thing. Now split the prize.




Winning team:

Votes:

I will cover the poster and reveal the parts in combination. Score what is on show out of ten.

| Maths | Picture | Story | The whole aid **10** |
|---|---|---|---|
| Maths + picture | Maths + story | Picture + story | |

Those seven numbers are the characteristic function $v$. Anchor the whole aid at ten and judge the rest against it.

**While you play**

In threes, for twelve minutes: the Theorist writes the maths, the Artist draws the picture, the Storyteller tells the story. One poster, three clearly separated parts. Then everybody votes once, for any aid but their own.

## Six ways the aid could have come together

| Order | Maths adds | Picture adds | Story adds |
|---|---|---|---|
| Maths, picture, story | | | |
| Maths, story, picture | | | |
| Picture, maths, story | | | |
| Picture, story, maths | | | |
| Story, maths, picture | | | |
| Story, picture, maths | | | |
| **Average** | | | |

Maths ____ Picture ____ Story ____

They should add to ____, which is the whole value of the aid. That property is called _____.

The three parts on their own are worth ____ + ____ + ____ = ____, so ____ of the aid's value came from the parts _____.

For each person, how much more did the Shapley value give them than their part was worth alone?

Maths Picture Story

The part that gains most is the one that _____, not the one that is best on its own.

What we just did in the room is **Question 1** on the Cooperative Games page, with the class's scores replaced by fixed ones.

| | | |
|---|---|---|
| **(a)** | Define a characteristic function game and the Shapley value. | [4] |
| **(b)** | Marginal contributions in each of the six orderings. | [5] |
| **(c)** | Hence the Shapley value. | [2] |
| **(d)** | Verify efficiency, and explain how the synergy is shared. | [4] |

vknight.org/gt/topics/cooperative-games.html

Every stage of today, and the part of the notes it was reading.

| What we did | Notes | In the textbook |
|---|---|---|
| Scoring every combination of parts | **18.2.1** | Cooperative Games: Definition: characteristic function game |
| The six orders of arrival | **18.2.9, 18.2.10** | Cooperative Games: Predecessors, Marginal Contribution |
| Averaging the contributions | **18.2.11** | Cooperative Games: Definition: Shapley Value |
| Why that split is the fair one | **18.2.5 to 18.2.8** | Cooperative Games: Efficiency, Null Player, Symmetry, Additivity |

vknight.org/gtb/chapters/cooperative-games

Quick quiz before you go: vknight.org/gt/topics/cooperative-games.html

Before the next class: read the Cooperative Games chapter as far as the Shapley value, and attempt Question 1.